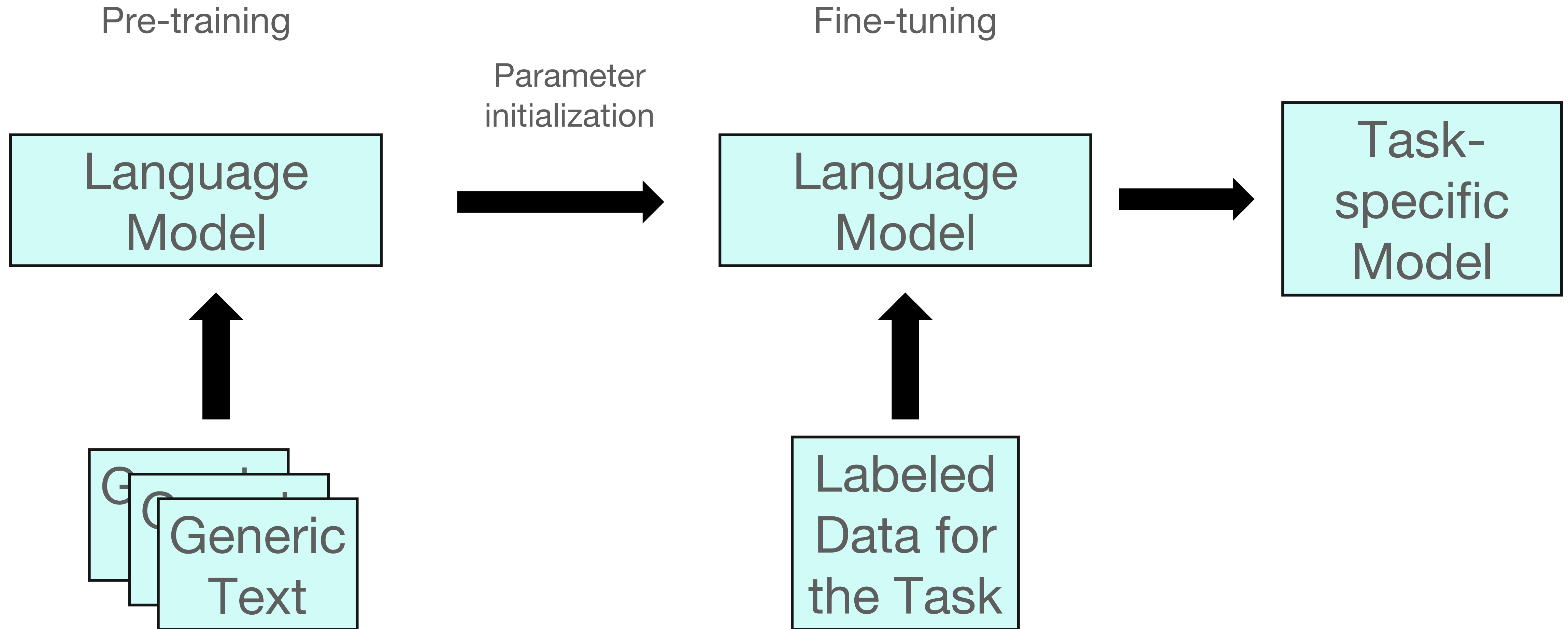
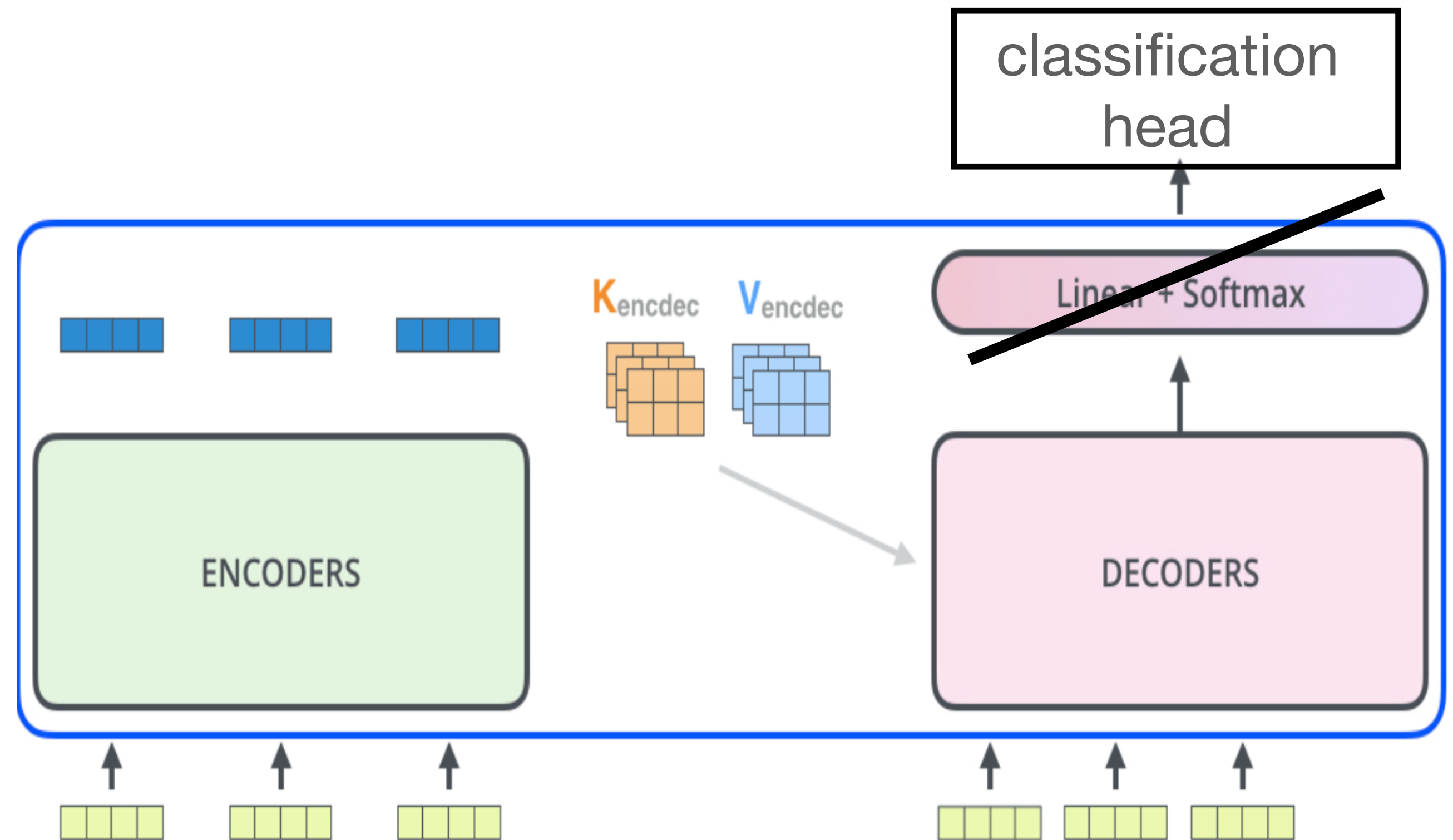


Pre-training and Fine-tuning



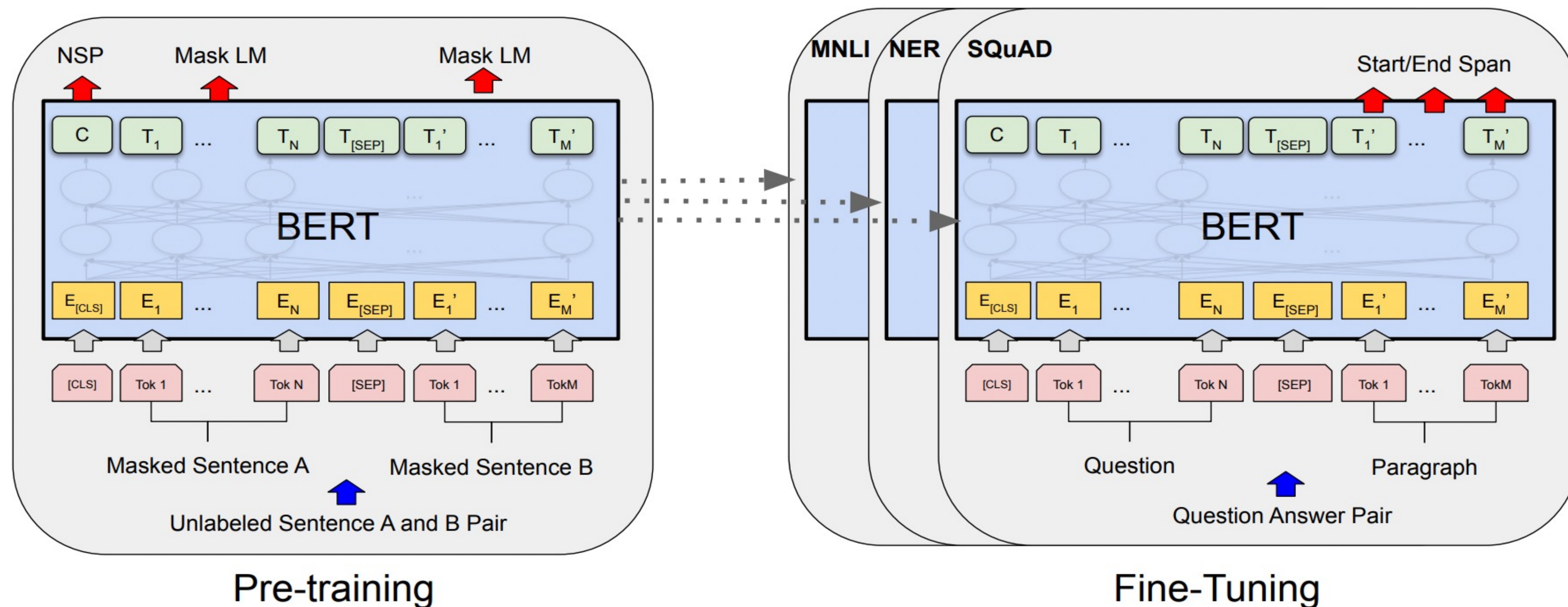
Pre-training and Fine-tuning

- For the fine-tuning task, we often replace the linear+softmax layer we used in the LM for decoding, with a task-specific sub-network
- Most common: a classification head – usually a feed forward network and a soft-max over the labels at the end



Fine Tuning BERT

- The standard paradigm for using BERT in NLP includes two phases:
 - Pre-training
 - Fine-tuning



Examples for fine-tuning BERT

- POS tagging:
 - Often viewed as an independent prediction task (not structured prediction)
 - A classification head for each word
 - Its input is the BERT embedding and its output is a soft-max in the dimension of the number of POS tags.
 - As usual, the j -th coordinate of the soft-max is interpreted as $P(y_j/x_1, \dots, x_n)$
 - Training set:
 - Each sample consists of the sentence $(x_1, \dots, x_n)$, index i , and gold POS tag for i -th token y_i^*
 - Cross entropy loss over $P(y_i^*/x_1, \dots, x_n)$

Examples for fine-tuning BERT

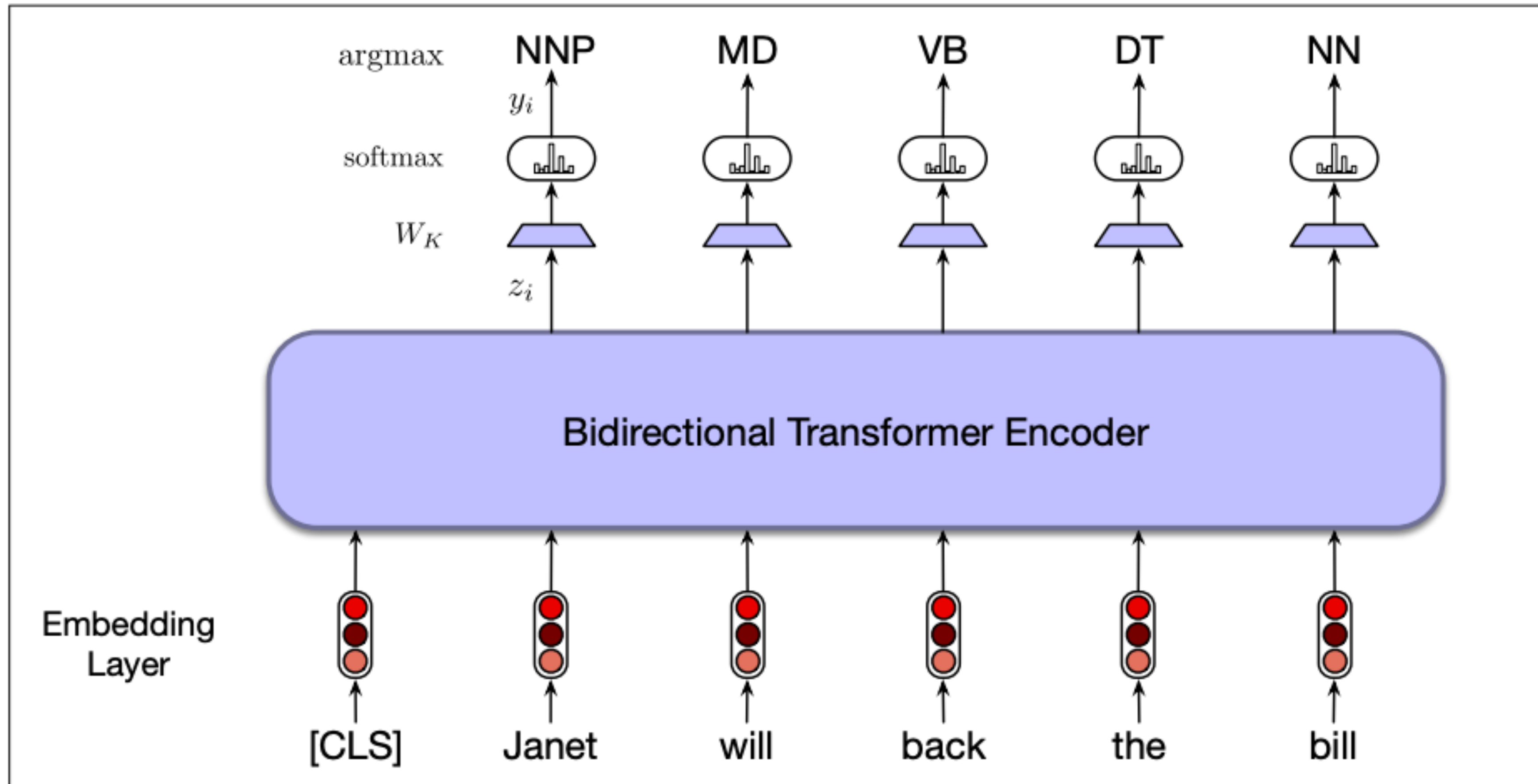


Figure 11.9 Sequence labeling for part-of-speech tagging with a bidirectional transformer encoder. The output vector for each input token is passed to a simple k-way classifier.

Next Sentence Prediction

- BERT is in fact pretrained not only as a masked language, but also as a model for Next Sentence Prediction (NSP)
- NSP is the task that requires, given two sentences, to determine whether the two sentences form a sensible consecutive pair or is just two random sentences
 - NSP training data: positive - pairs of adjacent sentences, negative - pairs of random sentences

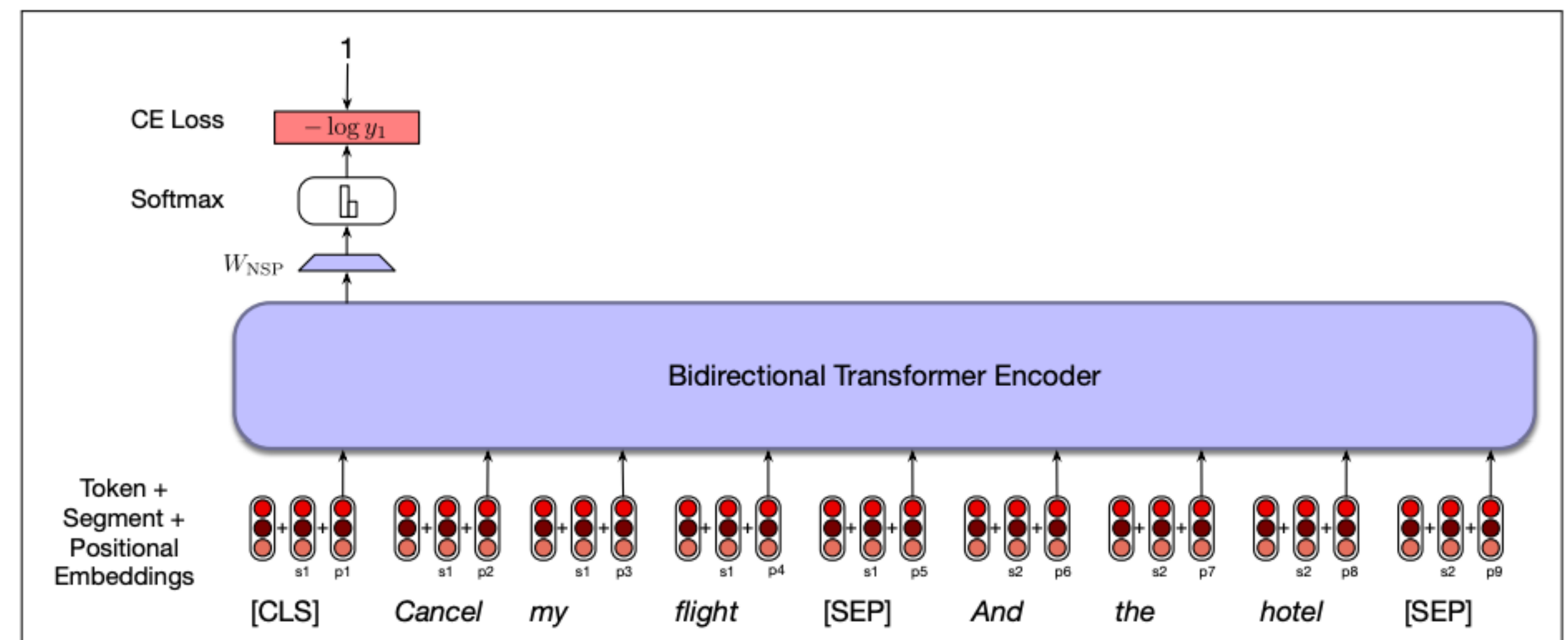


Figure 11.7 An example of the NSP loss calculation.

Next Sentence Prediction

- Each training sample is prepended with a special [CLS] token. A special [SEP] token is added between the sentences
- The loss function is over the [CLS] where the labels are 1/0
- [CLS] is often used for fine-tuning on sentence classification tasks

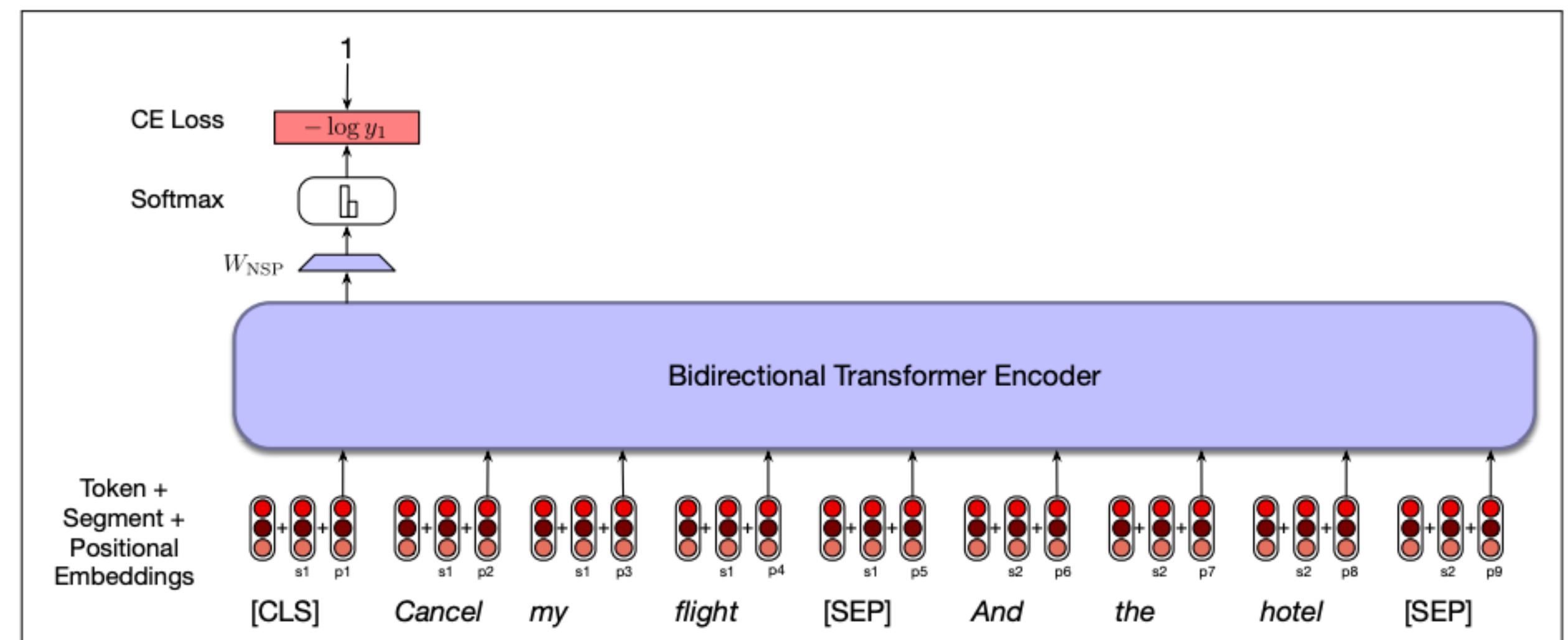


Figure 11.7 An example of the NSP loss calculation.

Examples for fine-tuning BERT

- Sentiment analysis:
 - Viewed as a sentence classification task
 - A classification head is placed over a special token ([CLS])
 - As before, usually a linear layer followed by a soft-max
 - The soft-max is interpreted as $P(class/x_1, \dots, x_n)$
 - *class* takes values in the set of potential sentiment classes
- Training set:
 - Each sample consists of the sentence $(x_1, \dots, x_n)$, gold POS tag for the sentence y^*
 - The loss function is the expectation over $\log P(y^* / x_1, \dots, x_n)$

Results of Fine-tuning BERT

- BERT achieved very impressive results across a range of token and sentence classification tasks, sequence prediction, as well as other baselines
- A very broad experimental setup in terms of task types and baselines
- It was widely adopted, and is still often used (about 5 years after its inception)
- For detailed results, see the original BERT paper

Fine-tuning Decoder-only Models

- Decoder-only models can also be fine-tuned
 - Implementation details vary but the crux is the same – using the parameters of the pretrained model as the initialization for a prediction model built on top of it

